



### Focus Mode for Chrome browser

A new feature tested on cutting edge development version of Google Chrome suggests the browser may get a Focus Mode soon. <https://dgit.in/mar19-56>



### Facebook's location privacy control

New location control for Android tries to give people more control over info they share on social media. <https://dgit.in/mar19-57>

necessary for our model generation as and re-run the process, we see a better decision tree. The exact path that the tree takes can be seen by clicking on the description tab.

So, if we have a training example with the regular attributes, we can, at least, determine the expectancy for survival by following the conditional paths. But hey, this can be a very tedious job if there are several examples for which the probabilities need to be found out, more so, in a decision tree with multitudinous branches. So, we shall have to automate the process by applying this model. Let's see how we can do that.

## APPLYING THE MODEL

We prepared a data set by mixing a few unlabelled examples with some labelled examples shown on the right.

Let's take this down step by step.

Retrieve the titanic-mixed data set and select the attributes we want to be present in our model which are as follows: the age, sex, no of siblings, no of parents and survived. Set the role for Survived to be labelled as we want to apply the model to get a prediction on this attribute

Here we use a multiply operator which basically creates two copies of the data that we have till this point. For one branch, we use the filter examples operator to filter out data with no missing survived attributes and for the other branch we also use filter examples to get an invert of the above data-set, i.e., data with missing survived values.

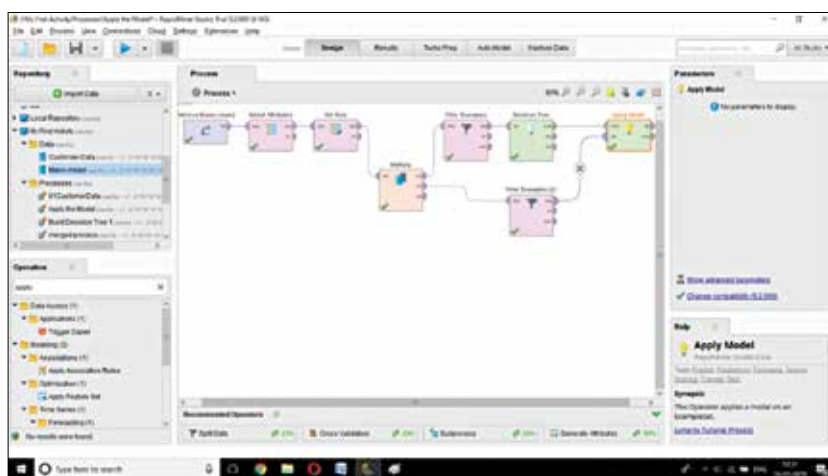
We had the decision table ready from the last demonstration so we are going to use the model output port of the decision table operator to connect it to the model input port of the apply model operator. The decision tree will essentially serve as the model.

For the second filter examples, we would want to connect the example output port to the unl (unlabelled) input port of the apply model operator.

For the output of the apply model operator, we wish to observe both the model and the labelled data-set and so



The improved decision tree



Our data set

we connect both of them to separate results set and run the process. Interpreting row no. 112 would be something like, "a male, aged 23 with no siblings or no parents is least likely to survive".

As a conclusion to this, let's test out the training performance of this model. For this, we need to tweak the process. Our goal would be to feed the training data (with filtered out examples containing no missing survived values) into the decision tree which forms the model for the input port of the apply model operator while simultaneously use a copy of the same filtered data directly into the unlabelled port of the apply model operator.

In essence, we are using the same data twice; once through the decision tree to see the prediction of the percentage of survival and another to directly obtain the results for what they are. Running the process we get to a survived column having definite yes' and no's and a prediction column which is a result of the decision tree. Now, we

can evaluate how good our model has performed. The model's result matches with the actual result for rows 1 through 4. However for row 5, there is a mismatch. In actuality, the person corresponding to row 5 had not survived, however, our model predicts an yes, thereby erroneously predicting so. Instead of manually finding out the anomalies, we can use the performance operator which does this calculation for us and gives us a result for the performance of the model.

Having said this, I'll have to say that this is only the tip of the ice-berg as far as the functionalities of Rapid Miner are concerned. It's indeed a vast tool and feel free to go through the docs at <https://dgit.in/RMiner>. RapidMiner has a free edition that limits you to 10000 rows of data and 1 logical processor at a time. However, you also get a 30-day trial for the Large edition. Paid editions start at \$2500 per user per year and have exclusive access to features like Auto Model and Turbo Prep.

Happy Rapid Mining! 🚀